But even more important than the time-horizon effect in Table 3.2 is the fact that the amount by which the shortfall risk is reduced depends on the correlation coefficient between the two assets that you are mixing in your portfolio. In other words, the lower the correlation, the lower the risk.

Compare, for example, the zero correlation case. In Table 3.1, looking at risk over a one-year period, the probability of shortfall was roughly 28%. In the case of a five-year horizon with zero correlation between the assets, the probability of shortfall is reduced to roughly 10%. Over 10 years, the same zero correlation between the two investments leads to a shortfall risk of approximately 3%. Isn't it remarkable what four (or nine) more years of investing can do?

The reduction is even more pronounced if you look at a –50% correlation. Remember, a negative correlation means that when one asset is doing better than average, the other is doing worse than average. In other words, they are generally moving in the opposite relative direction. The –50% is midpoint on a scale of –100% to 0% and measures the relative strength of this opposite movement. In Table 3.1, the one-year horizon case, the probability of shortfall is roughly 19%. In Table 3.2, the five-year horizon, the odds are reduced to a mere 2%. For 10 years, it is reduced to less than a quarter of a percentage point. Notice the effect of time and the effect of correlation.

What are the lessons from these observations? Two things will reduce the shortfall risk of your portfolio:

- The longer time horizon over which you hold the portfolio.
- The movement of the assets in your portfolio. The more independently they move, that is, the lower their correlation, the more your risk is reduced.

I like to argue that there are two dimensions to investment diversification—time and space. In other words, there are two aspects to diversification. By time, I mean the length of time you hold the portfolio—the longer you hold it, the more diversified your portfolio becomes. If you consider the essence of diversification to be the reduction of shortfall risk—owing to the imperfect correlation—then holding the portfolio for longer periods of time will also reduce shortfall risk.